

Assessing mackerel scad, *Decapterus macarellus*, in Cape Verde: Using a Bayesian approach to biomass dynamic modelling in a data-limited situation

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Abstract

The fisheries for mackerel scad, *Decapterus macarellus*, are particularly important in Cape Verde, constituting almost 40% of total catches at the peak of the fishery in 1997 and 1998 (≈3700 tonnes). Catches have been stable at a much lower level of about 2100 tonnes in recent years. Given the importance of mackerel scad in terms of catch weight and local food security, there is an urgent need for updated assessment. Stock assessment was carried out using a Bayesian approach to biomass dynamic modelling. In order to tackle the problem of a non-informative CPUE series, the intrinsic rate of increase, r , was estimated separately, and the ratio B_0/K , initial biomass relative to carrying capacity, was assumed based on available information. The results indicated that the current level of fishing is sustainable. The probability of collapse is low, particularly in the short-term, and it is likely that biomass may increase further above B_{msy} , indicating a healthy stock level. It would appear that it is relatively safe to increase catches even up to 4000 tonnes. However, the marginal posterior of r was almost identical to the prior, indicating that there is relatively low information content in CPUE. This was also the case in relation to B_0/K . There have been substantial increases in fishing efficiency, which have not been adequately captured by the measure used for effort (days or trips), implying that the results may be overly optimistic and should be considered preliminary.

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1. Introduction

Mackerel scad, *Decapterus macarellus*, is a particularly important species in Cape Verde, constituting almost 40% of total catches at the peak of the fishery in 1997 and 1998. However, this proportion has come down to about 20% in recent years, corresponding to 2104 tonnes in 2004 (Fig. 1). Pelagic species, namely tuna (*Katsuwonus pelamis* and *Thunnus albacares* primarily), various small pelagic species as well as wahoo (*Acanthocybium solandri*) dominate the fisheries catches in Cape Verde, accounting for between 70 and 75% of total catches. Concerning small pelagics, two other species, *Spicara melanurus* and *Selar crumenophthalmus*, have gained importance in recent years. Although these two species now constitute about half of total artisanal catches, they are of minor importance

in the industrial fishery. Most of the mackerel scad catches are taken with seine gears, both in the industrial and artisanal fisheries, with minor catches by artisanal handline.

The mackerel scad stock has been assessed on several occasions (Carvalho, 1993; Almada, 1997; Carvalho and Caramelo, 1999; Jardim, 1999), but these assessments need updating, including the application of more recent methodology. This is particularly urgent in the case of mackerel scad due to its importance in terms of catch weight and local food security.

Biomass dynamic models are one of the simplest analytical methods available that provide for a full fish stock assessment. Catch estimates and an abundance index for a number of years are the minimum requirements for this method, although ancillary data may be incorporated to improve parameter estimation. The logistic formulation in terms of biomass became known as the surplus production model (also called the Schaefer model; Schaefer, 1954, 1957), while the more recent term biomass dynamic models implies a dynamic fitting process (Hilborn and Walters, 1992; Haddon, 2001). The original method of assuming

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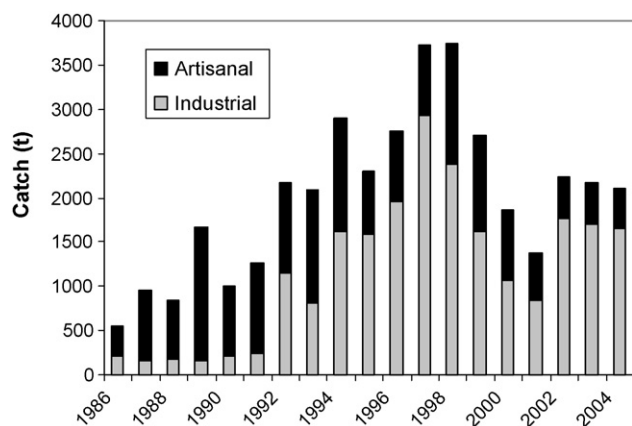


Fig. 1. Industrial and artisanal catches of mackerel scad in Cape Verde.

equilibrium conditions has been criticised for providing overly optimistic estimates of optimum effort and maximum sustainable yield (MSY) and has been associated with many failures in management (Botsford et al., 1997; Mace, 2001; Pauly et al., 2002). Equilibrium fitting methods, which assume a linear relationship between catch per unit of effort (CPUE) and effort, are now known to produce biased and unreliable parameter estimates and their use is not recommended (Hilborn and Walters, 1992; Haddon, 2001; Williams and Prager, 2002). On the other hand, the use of dynamic fitting methods and modern computers for parameter estimation has brought about a revival of biomass dynamic models, including estimates of uncertainty around parameters and risk assessment (Prager, 1994, 2002; McAllister and Kirkwood, 1998; McAllister et al., 2000, 2001; Abuanza et al., 2003; Carbonell and Azevedo, 2003; Mendoza and Larez, 2004).

One of the major difficulties in modelling fisheries data is the lack of contrast between fishing effort and stock abundance, which has been termed the “one-way trip” (Hilborn and Walters, 1992). This is observed as a steady decline of CPUE over time, which is very common in fisheries data sets. In the absence of ancillary information, an infinite number of plausible solutions exist in such a case, ranging from a large (biomass) stock with a low growth rate to a small stock with a high growth rate. All methods are expected to fail in this type of situation due to data failure (not model failure), including models using equilibrium and non-equilibrium fitting methods as well as age-structured models (Hilborn and Walters, 1992; Haddon, 2001).

Biomass dynamic models are usually considered a poor substitute for age-structured models, but they may be the only method available for carrying out a full stock assessment in data-limited situations such as in many tropical fisheries. Comparative studies of biomass dynamic and age-structured models have nevertheless shown that provided their main assumptions are satisfied, both types of models can perform equally well, giving the same answers for management. In some cases, biomass dynamic models may provide better estimates of management parameters and are thus more robust (Hilborn and Walters, 1992; Punt, 1995; Abuanza et al., 2003). Age-structured analysis is often not practical in tropical fisheries and the more simple

requirements of biomass dynamic models make them a cost-effective solution. Furthermore, it may be difficult to obtain species specific catches and CPUE in tropical fisheries, making it necessary to treat the entire catch (or part) as a biomass dynamic pool. Conventional age-structured analysis would not be appropriate in such cases.

We carried out an assessment of mackerel scad using a biomass dynamic model. In order to tackle the “one-way trip” problem, the intrinsic rate of increase, r , was estimated separately, and the ratio B_0/K was assumed based on available information. This study was inspired by recent developments in Bayesian stock assessment and its application in data-limited situations, which provide a theoretical framework for incorporating externally available information in stock assessment and thus reduce the uncertainty that is inherent to “one-way trip” situations (Punt and Hilborn, 1997; Hilborn and Liermann, 1998; McAllister and Kirkwood, 1998; McAllister et al., 2001). In this specific case, this is termed the prior knowledge of r and B_0/K specified as a probability density function. Uncertainty in parameter estimates and risk assessment of projections was addressed by using Bayesian techniques.

2. Materials and methods

In Cape Verde, fisheries data on catch and effort are available starting in 1981 with the implementation of a national statistical collection system for fisheries. As catch estimates are not considered reliable for the early years (Carlos Monteiro, INDP, personnel communications), we consider the time period from 1986 to 2001 in the present study. These catch and effort data were available at different levels of aggregation.

- Aggregated and raised yearly estimates for the period 1986–2002, available for artisanal and industrial fisheries separately.
- Aggregated and raised monthly estimates for the period 1996–2000, available by gear and island for artisanal and industrial fisheries.

CPUE is often used as an index of abundance in the absence of fishery-independent data such as the results of resource surveys. In such cases, CPUE is standardized (e.g. GLM) in order to remove the effects of different fleets and/or gears (e.g. Hilborn and Walters, 1992), but this was not possible in this study as the data available were highly aggregated. CPUE time series were available from the artisanal and industrial fisheries, which are based on effort measured in fishing trips (typically a day) and fishing days, respectively.

It is important to point out that artisanal fisheries in Cape Verde are multispecies and multigear fisheries, thus creating problems in determining directed effort to specific target species. The multivariate ordination method non-parametrical multi-dimensional scaling (MDS) was used to determine whether there are important spatio-temporal and gear effects to consider in artisanal CPUE data. Large pelagic and migratory species such as tuna and wahoo were eliminated from the data, resulting in a matrix of 216 samples, consisting of monthly CPUE data by

species/gear/island (1996–2000), and 32 variables (species). In order to facilitate interpretation, two additional factors, season and region, were defined based on the re-definition of month and island, respectively. The hot season was defined as from July to November. The region Barlavento (windward) included the northern islands (an arch spanning from S. Antão to Boavista). MDS was applied to a similarity matrix calculated with the Bray–Curtis coefficient (without data transformation), which has the desirable property of not being sensitive to joint absences or double zeros that was common in the data (zero catches for specific species) (Clarke and Warwick, 2001).

MDS is simple in concept, creating a sample map or ordination plot where the distances between points have the same rank order as the corresponding similarities between samples (Clarke and Warwick, 2001). MDS plots can be arbitrarily scaled or rotated, while maintaining the relative distance between points in two or three dimensions. A measure of goodness-of-fit is termed the stress value in MDS:

$$\text{Stress} = \sqrt{\frac{\sum_j \sum_k (d_{jk} - \hat{d}_{jk})^2}{\sum_j \sum_k d_{jk}^2}} \quad (1)$$

where $\hat{d}_{jk}$ is the predicted distance between the j th and k th sample points and the fitted non-parametric regression line corresponding to dissimilarity d_{jk} . Stress values larger than 0.3 indicate an almost random position of points; values lower than 0.1 correspond to a good ordination with clear interpretation; and values between 0.1 and 0.2 indicate a potentially useful ordination but interpretation should be handled with care (Clarke and Warwick, 2001). The PRIMER software was used for multivariate analysis (Clarke and Gorley, 2001).

3. Biomass dynamic model

The formulation of the Schaefer surplus production model in its continuous and discrete form is:

$$\frac{dB_t}{dt} = rB_t - \frac{r}{K}B_t^2 - C_t \quad \text{or} \quad B_{t+1} = B_t + rB_t - \frac{r}{K}B_t^2 - C_t \quad (2)$$

where r is the intrinsic rate of increase, K the carrying capacity, B_t the abundance (biomass), and C_t is the catch at time t (Hilborn and Walters, 1992; Haddon, 2001). In the Schaefer model, the biomass level that sustains the maximum sustainable yield (B_{msy}) is at one half of K . Maximum sustainable yield is defined as:

$$Y_{\text{MSY}} = \frac{rK}{4} \quad (3)$$

and the effort that sustains the maximum sustainable yield is:

$$f_{\text{MSY}} = \frac{r}{2q} \quad (4)$$

where q is the catchability coefficient, also called a nuisance or scaling parameter. One major assumption in the use of the surplus production model is that the catchability coefficient remains constant over time (Haddon, 2001).

4. Specifying priors

Preliminary attempts at modelling biomass dynamics of small pelagics in Cape Verde were unsatisfactory. Parameters of the Schaefer model were ill defined, resulting in large uncertainty around parameter estimates and reference points for management. This was due to lack of contrast in the CPUE time series or, in other words, “the one-way trip”. To overcome this limitation, r was estimated using the equation proposed by Sullivan (1991) for non-gadoid species. In this formulation r is a function of the von Bertalanffy growth coefficient (k) and of asymptotic weight (W_∞):

$$r = 0.947 + 1.189k - 0.095 \ln(W_\infty) \quad (5)$$

Mackerel scad growth parameter estimates were calculated as the geometric mean of available estimates ($k = 0.383$ and $W_\infty = 631$ g) (Appendix A). This gave a r point estimate of 0.790 per year and consequently, a lognormal probability density function (pdf) was defined (lognormal median = 0.790, lognormal mean = 1.079, lognormal S.D. = 0.790), corresponding to the uncertainty around this estimate. The value for standard deviation (S.D.) was chosen arbitrarily so as to correspond to a large coefficient of variation (CV) of 100%, following the recommendation by several authors to avoid an overly precise estimate that may be biased (Punt and Hilborn, 1997; McAllister and Kirkwood, 1998). This estimate of r and its probability density function (pdf) was used as the prior.

The prior of ratio B_0/K was assumed to have a mean of 0.8 and a standard deviation of 0.4 (normal distribution), based on the evolution of the fisheries in terms of catches, vessel types, and motorization. More details on this evolution are presented in the discussion.

5. Bayesian modelling

Bayesian surplus production (BSP) modelling followed McAllister and Babcock (2004). The BSP model uses a Bayesian parameter estimation method, in which the joint probability distribution of the parameters given the data (called the posterior distribution) is calculated from the likelihood of the data and the prior probability distribution of the data using Bayes theorem (McAllister et al., 1994; McAllister and Ianelli, 1997; McAllister and Kirkwood, 1998; McAllister et al., 2001):

$$P(\theta_i|y) = \frac{L(y|\theta_i)p(\theta_i)}{\int L(y|\theta_i)p(\theta_i)d\theta} \quad (6)$$

where $P(\theta_i|y)$ is the posterior distribution of the parameter vector θ_i given the data y , $p(\theta_i)$ is the prior distribution of the parameters and $L(y|\theta_i)$ is the likelihood of data given the parameters. When data contain little information, the posterior pdf tends to reflect the prior pdf. However, as data become more informative, $L(y|\theta_i)$ predominates and priors lose their influence over the posterior (McAllister et al., 1994).

The BSP model fits the Schaefer or Fletcher/Schaefer model to one or more time series of CPUE data. The CPUE data are assumed to be log-normally distributed about the model

predicted values, so that the log likelihood function of the CPUE data is:

$$\ln(L(y|\theta_i)) = - \sum_{j=1}^{nind} \sum_{t=iy_r}^{icur} J_t \frac{0.5}{\sigma_j^2} \left[\log \left(\frac{I_{j,t}}{q_j B_t} \right) \right]^2 - \log(\sigma_j) - 0.5 \log(2\pi) - \log(I_{j,t}) \quad (7)$$

where $I_{j,t}$ is the CPUE value for index j in year t , q_j the constant of proportionality or catchability coefficient for CPUE series j and σ_j^2 the variance in the lognormal likelihood function of data series j , $J_t = 1$ if $I_{j,t} > 0$, otherwise $J_t = 0$. Summations refers to the initial year (iyr), current year (icur), and the number of CPUE indices available (nind). In the present study, production modelling involves only one CPUE time series. The variance σ_j^2 can vary depending on the weighting method selected.

The BSP program approximates the posterior distribution using a numerical integration method called the sampling-importance resampling (SIR) algorithm. The SIR algorithm is based on the idea that the posterior distribution of a parameter of interest $g(\theta)$ can be reformulated so that it is defined in terms of its expected value ($E^{h(\theta)}$) with respect to an arbitrary probability density function $h(\theta)$, called the importance function (McAllister et al., 1994). The expected value can then be approximated by taking a large number (m) of draws from the importance function, so that the marginal posterior of $g(\theta)$ becomes:

$$E^{P(\theta|y)}(g(\theta)) \cong \frac{\sum_{k=1}^m g(\theta_k) w(\theta_k)}{\sum_{k=1}^m w(\theta_k)} \quad (8)$$

where

$$w(\theta_k) = \frac{L(y|\theta_k) p(\theta_k)}{h(\theta_k)} \quad (9)$$

is called the importance ratio or the weight of draw k .

In other words, the importance ratios form an approximation of the posterior distribution, placing on each θ_k the following mass:

$$F(\theta_k|y) = \frac{w(\theta_k)}{\sum_{k=1}^m w(\theta_k)} \quad (10)$$

The marginal distributions can be calculated directly from the importance draws, or a sub-sample can be taken to facilitate computation (the re-sampling part of the SIR).

For a sufficiently large number of draws Eq. (8) converges on the posterior distribution of $g(\theta)$ for any importance function $h(\theta)$, which allows a non-zero probability of sampling each point in the posterior distribution. In general, an importance function that closely approximates the posterior distribution will be more efficient, meaning that it will converge more rapidly to the posterior distribution. An importance function that places too little weight in the tails of the posterior distribution will be inefficient in approximating the posterior distribution. Details on diagnostics of convergence are available from McAllister and Ianelli (1997) and McAllister and Babcock (2004).

6. Risk analysis

Risk analysis involved a biomass projection of 15 years, using Eq. (1), under different management scenarios (TACs of 2000, 3000, 4000, 5000, 6000 and 7000 tonnes). This included the following steps: (i) randomly draw values for parameters from the posterior probability distribution (e.g. r and B_0/K in the surplus production model); (ii) using the drawn parameter values, project from the current year the model into the future and apply the policy of interest in each future year to predict its consequences; (iii) calculate the performance indices for each policy; (iv) repeat steps (i) to (iii) 5000 times; (v) produce a distribution of performance indices for the policy.

7. Results

MDS results indicated that there is a clear structure according to gear in the artisanal fishery (Fig. 2). Various plots considering the effect of other factors such as year, month, season, island, or region showed that these effects were not apparent (not shown). The resulting stress value of 0.17 indicates a potentially useful ordination and it is interesting to see a clearer grouping of the points concerning the seine fishery compared to other gears.

The calculation of the percentage contribution of each species to the similarity between gear groups can be used to define typical species in each fishery. In the case of the seine fishery, mackerel scad contributed with 94.1%, which indicates a fishery with a well-defined target species. However, there appears to be a tendency for increasing catches of small tuna (*Euthynnus alletteratus*) in recent years in this fishery (14% of total seine catches in 2000). The artisanal seine fishery for mackerel scad is rather small, comprising about 24 larger boats (≈ 10 m) with outboard engines, but characterised by high CPUE compared to other gears. These boats catch around 250 kg per trip in recent years, while the overall artisanal mean is 37 kg per trip. Most of the fishing activity takes place in the northern islands, where mackerel scad is more abundant. Nonetheless, the MDS results indicated that island or regional effects could be ignored, as the differences could be attributed to gear effects mainly (Fig. 3).

Concerning the industrial fishery, the fleet is composed of various types of vessels that vary in length from 7 to 22 m (closed deck and inboard engines). The main target species are mackerel

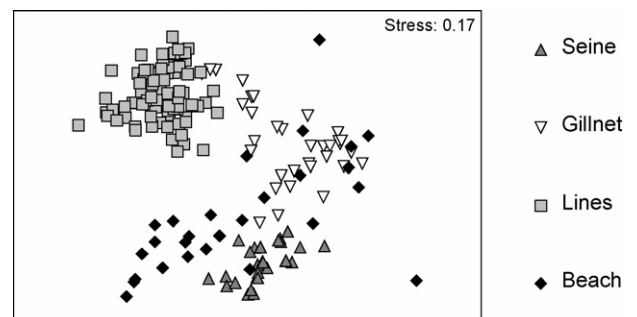


Fig. 2. Two-dimensional MDS plot based on the similarity of samples in terms of CPUE by species and grouped by gear. The symbols refer to the seine, gillnet, handline and beach seine fisheries.

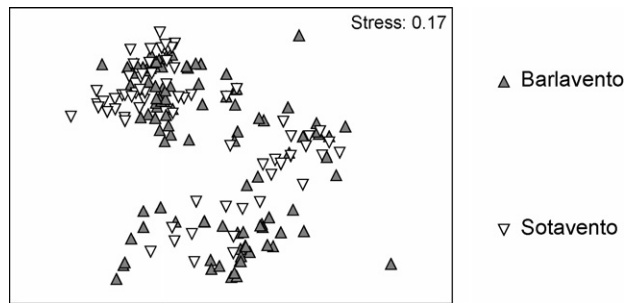


Fig. 3. Two-dimensional MDS plot based on the similarity of samples in terms of CPUE by species and grouped by region. Barlavento corresponds to the northern islands (S. Antao to Boavista) and Sotavento the southern islands.

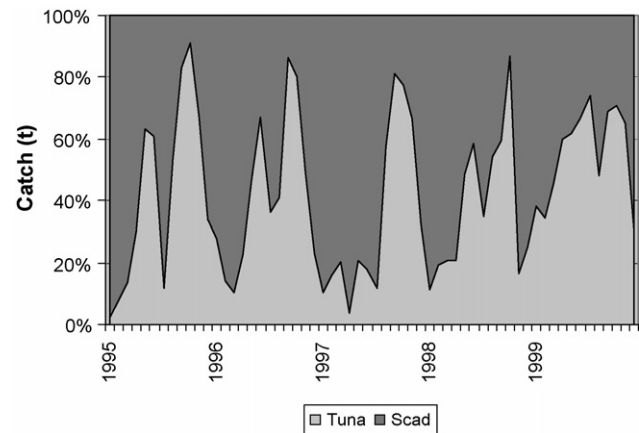


Fig. 4. Monthly catch composition of tuna and scad in the industrial fishery.

scad and tunas, skipjack tuna (*Katsuwonus pelamis*), in particular, as well as yellowfin tuna (*Thunnus albacares*). These fisheries are seasonal, where tunas tend to dominate towards the end of the year (Fig. 4). Various gears are used such as purse seine and pole-and-line as well as handline to a lesser extent. Disaggregated data on effort by gear was not available for the present study, so it was not possible to determine directed effort towards mackerel scad. Thus, CPUE was calculated as total mackerel scad catch by total effort.

CPUE time series were available from the artisanal and industrial fisheries, which are based on effort measured in fishing trips (typically a day) and fishing days, respectively. These effort units were assumed to be approximately the same (fishing days) and the two artisanal CPUEs (based on total trips and purse seine

trips) were scaled up in relation to the industrial CPUE (Table 1; Fig. 5).

Note that there is a general tendency for increasing in CPUE in the beginning of the time series. This appears to have been the result of increasing efficiency of vessels. In the industrial fleet, the jump to higher CPUE coincided with the introduction of the more efficient 11 m fibreglass boats. Furthermore, these higher CPUEs coincided with the arrival of two large Japanese freezer vessels that started buying mackerel scad in the early 1990s to be used elsewhere as bait for the Japanese fleet, creating an attractive market for this species. This market disappeared however in the late 1990s. In the artisanal fishery, the increase

Table 1
Catch, effort and CPUE of mackerel scad in Cape Verde. Overall CPUE is based on the average of industrial and artisanal seine CPUEs, but initial year in model estimation was set at 1992 (see text)

Year	Catch (tonnes)			Effort (days or trips)			CPUE (kg/day)			Scaled CPUE (kg/day)		Overall CPUE
	Total	Industrial	Artisanal	Industrial	Art total	Art seines	Industrial	Art total	Art seines	Art total	Art seines	
1986	544	208	335	2272	119851		91.7	2.8		197.5		197.5
1987	948	154	794	2218	134478		69.6	5.9		416.6		416.6
1988	841	175	665	1668	116993		105.0	5.7		401.4		401.4
1989	1671	166	1506	2301	154358		72.1	9.8		688.4		688.4
1990	1001	208	794	2021	137998		102.7	5.8		405.9		405.9
1991	1263	242	1021	2053	138534		118.0	7.4		520.3		520.3
1992	2169	1151	1018	1304	112737	5563	882.6	9.0	183.0	637.4	356.4	619.5
1993	2091	813	1278	1980	123016	4010	410.4	10.4	318.8	733.5	620.9	515.7
1994	2901	1623	1277	2829	134699	3639	573.8	9.5	351.0	669.2	683.5	628.7
1995	2301	1585	716	3106	128732	2621	510.2	5.6	273.2	392.6	532.1	521.2
1996	2759	1957	802	3426	134722	3908	571.2	6.0	205.2	420.1	399.6	485.4
1997	3725	2935	789	5057	140405	3149	580.5	5.6	250.7	396.7	488.1	534.3
1998	3740	2380	1360	5359	149022	5308	444.1	9.1	256.3	644.1	499.0	471.6
1999	2707	1622	1085	5949	155009	4332	272.7	7.0	250.4	493.9	487.6	380.1
2000	1864	1075	789	4822	173855	3705	223.0	4.5	212.9	320.2	414.5	318.8
2001	1379	849	530	3450	154408	3488	246.1	3.4	151.8	242.1	295.7	270.9
2002	2232	1759	473	5366	156741	3478	327.8	3.0	136.1	213.2	265.1	296.4
2003	2171	1695	476	n.a.	n.a.	n.a.						
2004	2104	1654	450	n.a.	n.a.	n.a.						
CPUE							458.4 ^a	6.5 ^a	235.4 ^a	70.6 ^b	1.9 ^b	

Scaling was calculated by multiplying with the specific scale factor relative to mean industrial CPUE (for example, scaled CPUE_(Art total) = 458.4/6.5 × CPUE_(Art total); scaled CPUE_(Art seines) = 458.4/235.4 × CPUE_(Art seines)).

^a Mean CPUE.

^b Scale factor.

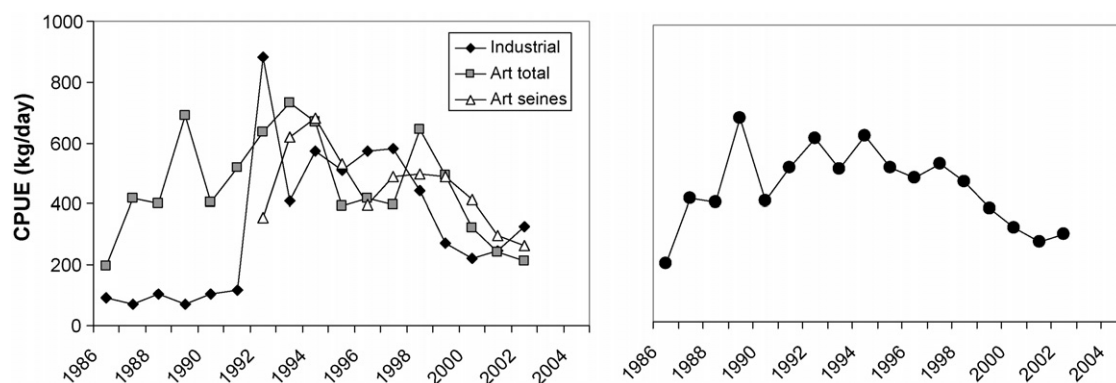


Fig. 5. Various CPUE time series available for mackerel scad (left-hand figure) and the estimated CPUE used as an index of abundance (right-hand figure).

in efficiency appears to be related to increasing motorization and this appears to stabilise around 1994 at around 70% (Fig. 6).

Based on the effects on catchability of for example the shift in targeting in the industrial fishery during the 1990s and the increasing efficiency of artisanal boats, it was considered necessary to restrict the CPUE time series to the period from 1992 to 2002. The overall index of abundance (CPUE) was estimated as the simple average of industrial CPUE and artisanal seine CPUE (scaled) for the time period (Fig. 5).

The results of the Bayesian surplus production model indicate that the stock of mackerel scad is being exploited sustainably

(Table 2). Recent catches of about 2100 tonnes are less than half the MSY estimate of 4700 tonnes MSY. Also, MSY is relatively well defined, although the CV is 53% (Fig. 7). The current level of biomass was estimated to be higher than B_{msy} by a factor 1.53 and the biomass level appears to be stable (Fig. 8). The coefficient of variation is generally high, which is to be expected, considering the priors for r and B_0/K .

It is important to point out that the marginal posterior for r is almost identical to the prior, indicating that there is relatively low information content in CPUE (Fig. 9). In other words, this is a typical “one-way trip” with a lack of contrast in the data.

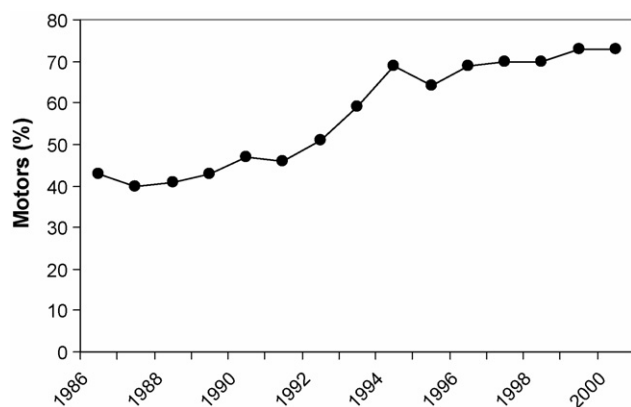


Fig. 6. Motorization of artisanal boats (outboard motors) (INDP, 2003).

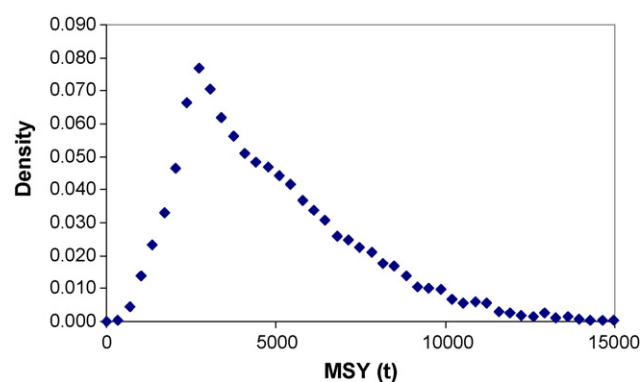


Fig. 7. The marginal posterior distribution of MSY.

Table 2
Estimated parameters of the Bayesian surplus production model (cur refers to current year in 2002)

	Mean	S.D.	CV
$E(K)$	27683	13483	0.49
$E(r)$	0.871	0.589	0.68
$E(MSY)$	4771	2528	0.53
$E(B_{cur})$	21105	12189	0.58
$E(B_{cur}/K)$	0.77	0.21	0.27
$E(B_0)$	22371	11619	0.52
$E(B_{cur}/B_0)$	0.97	0.32	0.33
$E(C_{cur}/MSY)$	0.60	0.39	0.65
$E(f_{cur}/f_{msy})$	0.76	2.70	3.56
$E(B_{cur}/B_{msy})$	1.53	0.41	0.27

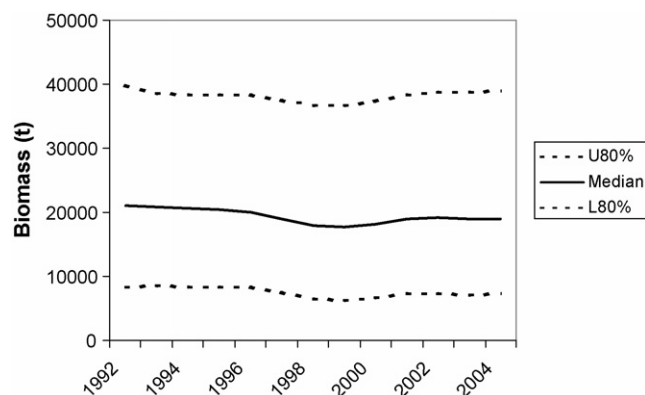


Fig. 8. Estimated biomass levels using the Bayesian model, showing median values and 80% confidence intervals.

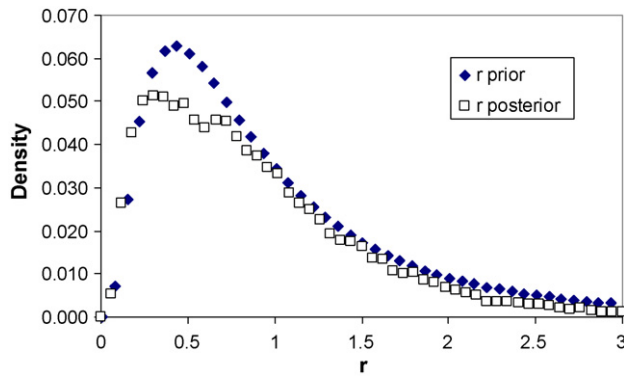


Fig. 9. The marginal posterior distribution and prior of r .

Note also that the model has estimated a B_0/K ratio of about 0.81, which is also identical to the prior.

Three performance measures were chosen for the purpose of decision analysis (Table 3). $B_{Fin}/K < 0.2$ is the probability that biomass at the end of the period (projections of 5, 10 or 15 year periods) is below 20% of K or, in other words, has collapsed; $B_{Fin} > B_{msy}$ is the probability that biomass at the end of the period is above the estimated biomass at Y_{MSY} or, in other words, that biomass is at healthy levels; $B_{Fin} > B_{2001}$ is the probability that biomass at the end of the period is greater than the biomass in 2001.

Decision analysis will depend on management objectives, but the results indicate that the current level of fishing is sustainable (≈ 2100 tonnes). The probability of collapse is low, particularly in the short-term, and it is likely that biomass may increase further above B_{msy} , thus indicating a healthy stock level. It would appear that it is relatively safe to increase catches even up to 4000 tonnes, depending on what is considered to be an acceptable risk.

8. Discussion

Previous assessments have been undertaken for the mackerel scad stock specifically (Carvalho, 1993; Almada, 1997; Carvalho and Caramelo, 1999; Jardim, 1999), but none of these studies deal with uncertainty in parameter or reference point estimates, although the study by Almada (1997) included risk assessment by projecting biomass forward with stochastic variability. More importantly, most of these assessments can be considered outdated, except for the study by Almada (1997), as the fisheries have evolved considerably since then. Note also that all of these studies included data available for the period 1981–1986, which was excluded from this study as these data have been found to be unreliable.

Jones length-based cohort analysis (Sparre and Venema, 1992) was applied on two occasions (Carvalho and Caramelo, 1999; Jardim, 1999), based on data up until the early 1990s. The Fox production model, which is the exponential formulation of the surplus production model, was applied separately to industrial and artisanal data for the same period (Jardim, 1999). These equilibrium based methods are nowadays considered inadequate by the scientific community.

The approach used by Almada (1997) included a much more thorough review and analysis of data, as well as various stock assessment models (length cohort analysis, age cohort analysis, and yield per recruit). Interestingly, cohort analysis was abandoned due to the quality of CPUE data for tuning purposes. The Beverton and Holt yield per recruit model was applied, which assumes equilibrium conditions. However, the coupling of a stock recruitment relationship to the model by Almada (1997) deals with the major limitation of the yield per recruit model. Almada (1997) estimated Y_{MSY} and biomass to be 5500 and 24,000 tonnes, respectively. The stock recruitment relationship used was highly uncertain, but it is nevertheless comforting that the results from this structurally different model

Table 3

Consequences of alternative TAC policies for the stock of mackerel scad in Cape Verde, projected from 2002 to 2017, using the Bayesian model

Horizon	TAC policy	$E(B_{Fin}/K)$	$E(B_{Fin}/B_{msy})$	$P(B_{Fin} < 0.2K)$	$P(B_{Fin} > B_{msy})$	$P(B_{Fin} > B_{cur})$
5-year	2000	0.77	1.54	0.06	0.88	0.87
	3000	0.67	1.34	0.1	0.8	0.02
	4000	0.55	1.1	0.21	0.67	0.01
	5000	0.44	0.88	0.33	0.53	0
	6000	0.35	0.69	0.44	0.4	0
	7000	0.27	0.53	0.54	0.29	0
10-year	2000	0.77	1.53	0.08	0.88	0.86
	3000	0.64	1.27	0.16	0.76	0.02
	4000	0.49	0.98	0.31	0.59	0.01
	5000	0.37	0.74	0.46	0.45	0
	6000	0.27	0.54	0.59	0.32	0
	7000	0.19	0.37	0.7	0.22	0
15-year	2000	0.77	1.53	0.09	0.88	0.87
	3000	0.62	1.24	0.19	0.75	0.02
	4000	0.46	0.93	0.37	0.56	0.01
	5000	0.34	0.68	0.52	0.42	0
	6000	0.24	0.48	0.65	0.3	0
	7000	0.16	0.33	0.76	0.2	0

The expected values (E) of the performance measures are given as well as the probabilities (P) (see text for explanations).

are comparable to this study in terms of Y_{MSY} and biomass, when considering the uncertainty involved.

It is evident from our study that considerable care should be taken in assessing the quality of data available and that knowledge of the fishery is essential. Structurally different models should in principle give the same answers, given that the requisites are not violated. Surplus production models have been heavily criticised, most notably by Larkin (1977 in Mace, 2001), but a revival has occurred under the new name of biomass dynamic models. The application of the surplus production model to non-informative data was the main reason for these failures (Hilborn and Walters, 1992). An attempt was made in the present study to estimate the parameters of the production model using bootstrap techniques, but this failed due to the nature of the data. In such cases, it becomes particularly relevant to use Bayesian methods, where we expect r to be within a definite range (prior knowledge). This r parameter was estimated using an empirical equation and assuming a variation based on other studies, which gave reasonable estimates, considering that these are small, fast-growing species in tropical waters.

In relation to B_0/K , this was assumed to be 0.8, based on various survey reports. The latest acoustic survey was carried out in 1997 and gave an estimate of about 15,000 tonnes of small pelagics, mostly *Decapterus macarellus* (Marques et al., 1997). Regarding the other acoustic survey undertaken in 1981 (Stroemme et al., 1982), it appears most likely that the estimate of 50,000–65,000 tonnes of small pelagics biomass was exaggerated and inaccurate, but it suggests that the stock was close to virgin levels at the time (e.g. Stroemme et al., 1982; Magnússon and Magnússon, 1985). In both acoustic surveys there were problems of accessing the area, as the continental shelves are generally narrow around the islands. Considering that small artisanal boats took most of the mackerel scad catches in the 1980s, a biomass level of around 0.8 of K , together with a wide CV of 40%, appears a reasonable assumption at the start of the CPUE time series (1986). The targeting of mackerel scad by the industrial fisheries is a later development, which started in the early 1990s.

The approach used of defining priors was an oversimplification, but more data intensive methods such as the demographic or meta-analytical approaches can easily be implemented in a formal assessment setting. Nonetheless, we consider an in-depth analysis of available non-aggregated data a more important issue in order to study changes in catchability (q), directed effort in the industrial fishery, and better standardisation procedures for CPUE. A modelling study using Ecopath with Ecosim showed that in order to obtain reasonable simulation results concerning catches and CPUE of major species, it was necessary to assume that relative fishing rate almost tripled over the time period from 1986 to 2000 (Stobberup et al., 2004). In the case of small pelagics, consisting mostly of mackerel scad, relative fishing rate was assumed to have increased by a factor 5. This indicates that there may have been substantial increases in efficiency, which has not been captured by the measure used for effort (days or trips). Thus, the results given may be overly optimistic and the estimate of Y_{MSY} should be considered preliminary, as well as the probabilities of the consequences of the various TAC policies.

There is much debate on the issue of priors, but the growing consensus appears to favour Bayesian approaches provided that (i) they are properly implemented, (ii) informative priors are based on real independent information, and (iii) the sensitivity of results to so-called non-informative priors is investigated properly. If the data is poor and non-informative, the posterior distribution will follow closely the prior distribution, which emphasises that care should be taken in defining priors. This was the case in the present study, where the posteriors of both r and B_0/K closely followed the prior, thus the danger of defining overly precise estimates that may be highly biased.

It is important to note that in order to calculate the probability of the hypothesis one needs to apply Bayes theorem, which is mathematically proven and widely accepted (Punt and Hilborn, 2001). This has important theoretical consequences in the application of decision theory, which requires probabilities. The sampling distributions of estimates generated by for example bootstrapping do not represent probabilities, but they have often been applied and interpreted as if they were probability distributions for alternative hypotheses. Nonetheless, the use of bootstrap distributions in the evaluation of alternative harvest policy has provoked controversy (McAllister and Kirkwood, 1998; Punt and Hilborn, 2001).

One of main weaknesses in the management process has been termed the ratchet effect (Botsford et al., 1997). When under political pressure, managers will allow harvests to increase for their short-term benefits to society (jobs and profits) when fishery scientists cannot specify with certainty that the next increase will lead to over-fishing and collapse. Until recently, policy advice from stock assessments has typically been based on point estimates of management reference points, not explicitly accounting for uncertainty. When facing conflicting interests such as high biological yields and high rates of employment and profits, it appears that these reference points were often not taken seriously. Thus, much emphasis has been placed on uncertainty and risk analysis in recent developments of stock assessment methods (Punt and Hilborn, 1997; Sainsbury, 1998).

The Bayesian statistical approach offers an elegant and theoretically consistent framework within which to carry out stock assessment and provide policy advice (McAllister and Kirkwood, 1998). Bayesian statistics also provide a theoretical framework that can deal with relatively simple data-limited situations as well as the consideration of structurally different models as alternative hypotheses.

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Appendix A

Sources of parameters used in the estimation of the intrinsic rate of increase (r) with the equation by Sullivan (1991)

Reference	Von Bertalanffy parameters				Length/weight rel. (Almada, 1997)			Iintrinsic rate of increase (<i>r</i>), Sullivan's equation
	Linf	<i>K</i>	Geomean		<i>a</i>	<i>b</i>	Calculated Winf	
			Linf	<i>K</i>				
Carvalho (1993)	34.0	0.49	35.3	0.38	0.0078	3.17	631	0.790
Jardim (1999)	31.5	0.43						
Carvalho and Caramelo (1999)	41.0	0.30						
Almada (1997)	30.1	0.34						
Fishbase (http://filaman.ifm-geomar.de/home.htm)	41.7							

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